

Language Detection Transcription Translation Tool (LDT3)

“Our goal is to aide learning languages and being able to seamlessly convert real time audio into a target languages text by using a combination of powerful models alongside a unique website design that will allow users to be creative and learn languages on a path that they create themselves. The problem with current language tools is their limiting linear learning style which enforces a straight path for all users and set vocabulary for each stage, whereas for our LDT3 website will allow for users to learn what they want in a fluid manner based off their needs and interests at the moment with capabilities to save different journals of learning for different occasions and languages. As I will code for accuracy of the models, blending them together to create a functional detection, transcription, and translation effect while designing the website to fit my user goals. I will have my mentor proceed to check up on progress weekly, give insight to better tools to use and ideas to make it a more unique and powerful tool, and assist with resources or any issues I am facing while creating this project. Overall this tool will be a useful and fun way to learn languages with other smaller tools that will be embedded into it for extra functionality.

Josh King, Project Manager

Mentor: Jialu Li

February 17th 2026

Revision History Table		2/17/2026
Version	Summary of Changes	Date
V1	Initial Information listed – First page, revision table	2/17/2026

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1. Executive Summary

The Executive Summary was written by Josh King

LDT3 is a dynamic language learning tool designed to stray away from common linear learning and focus more on the user's goals and own creativity. It also doubles as a powerful transcription tool that can help convert your real time audio into live accurate transcription text which will also be translated to any of the available languages listed. With the ability to speak or type manually, this website will automatically convert that input into an accurate translation for a chosen language, with a journal like design where the left page is your source language (in most cases will be English) and the right page will be the translation. This style of transcribing is great to be able to compare what you inputted originally and see what the translated output should be which can help with associating your base language to the target language words and with sentence structure. Since it is a multifaceted tool it also allows for quick text conversions for tasks where you need to switch languages accurately quickly, along with the capability to stream audio and put out real time transcriptions for user interactivity.

This product is to not replace, but bridge the gap that other language tools are missing. Popular apps such as Duolingo or Anki use a linear learning approach that are great initially to understand the basics and to learn random vocabulary slowly but get to a point where all you know how to do is order a coffee and say a couple other random statements/questions that probably will not be applicable for your trip to Japan in the next month. As a complement to existing juggernaut applications such as those listed, LDT3 aims to allow the user to take control of the situation and learn what they want to learn more specifically by allowing them to talk to their computer and receive immediate transcriptions of both their language and target language which will allow them to learn dynamically. The ability to save these bilingual journals and title them into categories gives the users the freedom to create logical learning sessions for themselves and things to study of their choosing, with the ability to turn the journal into a one-way card to help study harder after its been created and saved! As mentioned this tool as dual purposes as a simple translator and transcriber as well which would also help those who are in content creation for captions, or for writing papers in another language, etc.

The process will be first to create the logic behind all the functionalities of the tool described with a python file containing a detection model to know what language is being spoken, a transcription model that will accurately transcribe the audio to the native text and lastly a translation mode that will take the transcription models output as input and give an accurate translation. It is most important to make sure these are working smoothly and accurately before proceeding. With that in mind using open sourced models for speed, efficiency and most importantly accuracy will be necessary such as OpenAI's Whisper model for transcription and Meta's NLLB for translation with some tuning to adjust for our project. A home made Convolutional Neural Network will be tested for detection or from Whisper as well. The key difficulty will be setting each of these models up and having them work seamlessly together. For real time streaming audio Kyutai Labs models will be used to low latency transcriptions. After all logic is built, designing the website will be the next crucial step. Adding functionality and connecting it to the Jetstream2 GPU allocation and hosting it online will be the last major hurdle but after that the whole project will be functional for use!

Team Member	Responsibility
Josh King	All

2. User/Market research

As stated in the Executive Summary, this tool is supposed to be a complement to other language learning tools such as Duolingo and Anki and not be a complete rival. Other language learning tools have set paths that are linear and begin feeling forced with the vocabulary you are stuck learning though do help with the initial comfortability with the language however, LDT3 aims to allow users that are becoming increasingly comfortable to a new language take control and get creative with learning what they want to learn. This fluid self-set limit approach to learning takes dedicated users that seek to learn seriously and have the creativity needed to create their own customized path of learning by being able to speak about what they want and save these bilingual journals into categories for later studying along with the ability to test yourself once they are saved by hiding the other language in study mode. The US Census claims only around 10-20% of US citizens can speak another language and as for our market, Duolingo, has around 50 Million daily users active. That is around 15% of the population of the US though Duolingo is available to most countries, meaning this market is alive and active! For all language learning apps, the estimation for total usage is around 750 Million people globally with 2/3rds of users preferring free apps supported by ads.

Duolingo was reported to make \$1 billion in revenue in 2025 alone while other places like Rosetta Stone (changed ownership) reported around \$30 million last year and Babbel a Berlin based subscription platform earned 330 million euros. From what we know of their user bases and the global user base, I believe this market has space and activity to support a complimentary language tool. It would be most effective to make the tool free to use and have ads in place to generate income based on webpage visits to comply with the 2/3 majority of users preferring free tools instead of a subscription plan.

User feedback about language tools seem to center around the rigid learning path of learning seemingly random things for the majority of the time. It helps originally by getting comfortable with basic words and how to start using phrases the correct way with the addition of adding to your vocabulary but after a long period of time you realize that with the current tools you can only say a couple niche things with extra vocabulary that you won't be able to use otherwise in that scenario. They want a way to have that nice introduction of basics but also a way to go off on their own and specify the types of things they learn to fit their own path. LDT3 aims to allow for the creative mind to discipline themselves and learn what they want by letting the user speak and create their own structured journals to test themselves against later on. With categories and language tags to keep them organized!

3. Product Features

MINIMUM FEATURES

Feature 1: Journal

This section was written by Josh King

The journal is the main component of the website as it will be where the models built to transcribe and translate data showcase the outputs and will be used for learning. This journal will be of 2 pages (left and right) with the left page being the source language (audio or manually typed) and the right page being the translated text. It will have the ability to be titled for organization and saved to go back and look at.

Feature 2: Study Mode

This section was written by Josh King

Study mode is a feature embedded within the journal feature, once a journal is saved you can go into study mode which makes it a flashcard and allows for you to test yourself on the journal you saved.

Feature 3: Real Time Audio Transcription

This section was written by Josh King

Above the journal will be a real time audio transcriber that is captioning your speech with low latency in batches, this will register below the journal below as the journal will be a slower more accurate model to ensure translations are precise. The Real Time audio is for showcasing the ability to capture audio and rapidly give text outputs back to improve accessibility and enable a more conversational style of tool.

<i>Iteration</i>	<i>Transcribed (seconds)</i>	<i>Transcribed GPU (seconds)</i>	<i>WER Score</i>

Feature 4: Cross Language Usage

This section was written by Josh King

The ability to switch languages and use any target language with any source language, with buttons indicating which languages are available and which to translate to.

<i>Iteration</i>	<i>BLEU Score</i>

ADDITIONAL FEATURES IF TIME PERMITS

Feature 1A: Romanization of Languages

This section was written by Josh King

For languages that do not share our alphabet, adding the Romanization of other translations below their native characters so it is more accessible for users to read/pronounce.

Feature 1B: Text-To-Speech Capabilities

This section was written by Josh King

The ability to listen playback audio to listen to the transcriptions (mainly English, maybe of other languages) the goal would be for accessibility and to hear the pronunciation of words. It also would enable conversational like learning.

Feature 1C: Automatic Video Captioning

This section was written by Josh King

The ability to upload a video and have its audio be auto captioned in any available language if the video requirements are aligned.

4. Project Timeline & Gantt Chart

Milestone	Date
Initialized GitHub Repository	12/23/2025
Audio Recording Capabilities	1/2/2026
CNN Detection Built/Transcription Model	1/8/2026
Mentor – Jialu Li	1/30/2026
Testing Moshi Realtime audio	2/5/2026
Translation Model Added + Tied all functions together	2/10/2026
GitHub Repository Restructured + bug fixes	2/12/2026
Initial Website Design + Proposal	2/17/2026

Table 1: Milestone Schedule

5. Ethics

#	Question	Generally	Data Breach
1	Could a user sell drugs or other illegal items on your platform?	Y/N/M	Y/N/M
2	Could a user of your platform engage in sex trafficking?	Y/N/M	Y/N/M
3	Could a user sell class notes or cheat on their homework on your platform?	Y/N/M	Y/N/M
4	Could a stalker use your project to find someone?	Y/N/M	Y/N/M
5	Could your app be used to spy on or track individuals?	Y/N/M	Y/N/M
6	Could your app/software access the camera or microphone and record things without users being aware?	Y/N/M	Y/N/M
7	If someone uses your platform, could they be re-traumatized or have their mental health impacted in some way?	Y/N/M	Y/N/M
8	Could your algorithm promote material that would traumatize or upset individuals?	Y/N/M	Y/N/M
9	Would your users be upset if the data you collect was given to someone else?	Y/N/M	Y/N/M
10	Could a data leak potentially lead to identity theft?	Y/N/M	Y/N/M

11	If your site was hacked, would users of that product potentially lose their job, spouse, or family?	Y/N/M	Y/N/M
12	Should there be an age limitation on your product?	Y/N/M	Y/N/M
13	Could someone use your product to find, contact, and potentially commit elder abuse?	Y/N/M	Y/N/M
14	If the data on your platform was breached, could it be used to blackmail the users?	Y/N/M	Y/N/M
15	Does the existence of your project imply that a particular racial group, gender, religion or other protected category is inherently bad, gross, or unwanted?	Y/N/M	Y/N/M
16	Could your product be used to commit hate crimes against a specific group?	Y/N/M	Y/N/M
17	Does the primary content of your game or algorithm focus on something considered deeply unethical?	Y/N/M	Y/N/M
18	Does your game or software contain race, gender, or other stereotypes?	Y/N/M	Y/N/M
19	Could users of your app scam other individuals?	Y/N/M	Y/N/M
20	Is your particular algorithm biased towards predicting correctly only for one race, gender, or other group?	Y/N/M	Y/N/M
21	Are the users of your project, players of your game, or those being surveyed for your data aware of how their data will be used?	Y/N/M	Y/N/M
22	What are the possible misinterpretations of your results? For example - would a white supremacist or misogynist be stoked about your results if they misinterpreted it?	Y/N/M	Y/N/M
23	Does the use or purchase of your data potentially contribute to a dangerous group or regime?	Y/N/M	Y/N/M
24	Could your virtual reality environment cause injury to the user?	Y/N/M	Y/N/M

- Are your study participants or game players aware that their data will be collected and used? **Y/N/M** **Y/N/M**
- 25 Does your game or app contain addictive design elements without benefit to the user? **Y/N/M** **Y/N/M**
- 26 Does your survey contain an aspect of compulsion or unusually large incentive, that would command users to take it even if it was to their detriment? **Y/N/M** **Y/N/M**
- 27
- 28 Could your research outcomes harm an individual or entity? **Y/N/M** **Y/N/M**

6. Approvals

The signatures of the people below indicate an understanding of the purpose and content of this document by those signing it. By signing this document, you indicate that you approve of the proposed project outlined in this Statement of Work, the division of work, the Ground Rules and that the next steps may be taken to create a Product Specification and proceed with the project.

This document is based upon and supersedes the *Project Proposal_LDT3_V1* Deviations, (versus clarifications), from the PDR have been clearly noted. For any requirements not listed in this SOW, the PRD requirements shall remain in effect.

Approver Name	Title	Signature	Date
Josh King	Team Project Manager	<i>Josh King</i>	2/19/2026
Josh King	Project Manager	<i>Josh King</i>	2/19/2026
Jialu Li	Advisor	<i>Jialu Li</i>	2/19/2026
Nitika Sharma	Instructor		

Additions

Section	Author	Word Count
1. Introduction	Josh King	232
2. Executive Summary	Josh King	622
3. User/Market Research	Josh King	497
4. Product Features	Josh King	392

7. Appendix

A. Advisor Engagement

1) Project Team Responsibilities

- The Project Manager will set up and facilitate a weekly call/meeting with the Faculty Advisor. The Project Team will provide weekly status updates to the Faculty Advisor including upcoming deliverables, critical issues, and any adjustments to the Project Plan.
- Documents will be provided to the Faculty Advisor with adequate time for review and signature. The time necessary for review will be agreed with the Advisor. The minimum review time will be 3 days prior to the document due date.
- Design files will be provided to the Faculty Advisor as requested in a format agreed to with the Advisor.
- Support requirements will be clearly requested from the Faculty Advisor with the dates required and an adequate time for fulfilling the request.
- Modifications requests to the Project Plan by Faculty Advisor will be reviewed and agreed to within 1 week of the request.

2) Faculty Advisor Responsibilities

- The Faculty Advisor will provide knowledge and expertise to help the group stretch their skills.
- The Faculty Advisor will participate in a weekly or bi-weekly call/meeting with the Project Team to review the project status, upcoming deliverables, priorities, issues, and progress to the agreed Project Plan.
- The Faculty Advisor will provide document review, feedback and approval, rejection, approval with contingencies with adequate time for the Project Team to meet the course due dates.
- The Faculty Advisor will provide feedback to requested support requirements from the Project Team. This includes feedback and guidance on design implementations decisions, design files, test plans, test procedures and test results.
- The Faculty Advisor shall provide technical advice and guidance to the Project Team answering inquiries approximately 1 hour per week.
- Modifications to the Project Plan by the Project Team will be resolved and documented within 1 week of the request.
- Grade the finalized project using a skill-based rubric
- Attend iShowcase in May.

B. Ground Rules

As a team and as individual team members, we agree to:

1. Stay focused on our objectives and goals.

Each time the team meets, we will clearly define our objectives and desired outcomes at the beginning of the meeting. We will politely remind team members if we are getting off track.

2. “Sidebar” any issues that are relevant but not consistent with the immediate objectives.

Occasionally, important matters are raised that are not relevant to the immediate goals of the meeting. To keep the group on track, but avoid losing the issue, create a “sidebar” where these topics can be listed and discussed later.

3. Listen when others are speaking.

We will listen and consider others’ input before adding our own comments.

4. All viewpoints will have an opportunity to be heard.

We understand that some team members may be quieter than others. We will make an effort to get each team member’s viewpoint and that no one dominates the discussion.

5. Differences of opinion will be discussed respectfully

We will identify areas of agreement before assessing areas of disagreement. We will encourage each other to look beyond our own point of view. We will discuss different ideas respectfully. As a team, we will weigh the merits of different opinions and agree on a process for choosing a direction. All team members will respect and follow the decision or direction.

6. Look for the good points in new ideas.

We will endeavor to explore the value in each idea as we assess and select our path forward.

7. Focus on the future, not the past.

We will use our past experience to inform our decisions, but focus the discussion on the future objectives. Blame for past performance is counterproductive, we will focus on finding solutions.

8. Agree upon specific action items and next steps.

At the end of each meeting and discussion, we will summarize and agree on specific next steps, action items and assignments.

9. Accountability

As team members, we will each be responsible for our individual assignments and contribution to achieving the team objectives and goals. We will honor our responsibilities and not let our team members down.